Hamza Rashid

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EDUCATION

University of Georgia, College of Engineering – Athens, GA — Aug. 2026 – Dec. 2029

B.S. in Electrical and Electronics Engineering

Relevant Coursework: Physics, Intro to Electrical & Computer Systems Engineering, Biometric Systems

TECHNICAL SKILLS

Hardware: Circuit design & analysis, soldering, breadboarding, PCB layout, oscilloscopes, Raspberry Pi, Arduino
Software: Python, OpenCV, Flask, Multisim, Fusion 360, Wireshark, I2C/SPI/UART/GPIO interfacing
Other: Bengali (fluent), Spanish (limited working proficiency), Google AI Professional Certificate (2026)

PROJECT EXPERIENCE

AgriGuard: Autonomous Pest-Deterrent System, IEEE FYC Team Project – Athens, GA — Sept. 2026 – Present

- Collaborating in a team of **4** to develop a Raspberry Pi and OpenCV vision pipeline (**15 FPS**, targeting **85%** detection precision under controlled lighting) that drives **2** pan-and-tilt servos via an I2C PWM controller, aligning the spray nozzle within **5**° of each detected target across the camera's full field of view
- Using a nozzle-mounted time-of-flight sensor to range targets from **0.5 to 1.5 m** (targeting distance error under **5 cm**) before switching a GPIO-controlled relay to fire the DC water pump in ∼**500 ms** bursts, targeting **70 of 100** successful hits across repeated test runs to protect crops from pests through deterrence

Credit Scoring Platform for Egypt, KSU Fintech Hackathon – Kennesaw, GA — Mar. 2026

- Placed **2nd of 40 teams** building a proof-of-concept fintech platform (Python, Flask, Pandas) that pairs e-wallet data with secure government ID verification to assess credit access for historically underbanked users in Egypt
- Integrated the live Egypt National ID API (Arya.ai) for identity verification and OpenCV with DeepFace for facial matching against the ID photo; credit scoring was a theoretical, rules-based demo, with encryption and KYC compliance scoped as required hardening steps for a production deployment

VEX Disc-Golf Launch Robot, VEX Robotics, Technology Student Association – Martinez, GA — Fall 2022

- Designed and wired the robot's flywheel launcher subsystem, driving **2** 393 motors through VEX Motor Controller 29s off a **7.2V** NiMH battery pack to generate consistent, long-range disc launch velocity
- Programmed flywheel speed control in VEXcode using a quadrature shaft encoder for closed-loop RPM feedback, reducing shot-distance variance and placing **2nd in state** against competitive teams from across Georgia

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant, UGA Cultivate Lab – Athens, GA — Oct. 2026 – Present

- Shaping curriculum design for ECSE 1100 (Intro to Electrical & Computer Systems) under Dr. Sarah Jane Bork, impacting **350+** students annually by integrating AI and, with **2** other researchers, redesigning **3** lectures, **2** labs, and assessments

SAT Tutor, Independent / Schoolhouse.world (Khan Academy) – Remote — 2025 – Present

- Tutored students in SAT prep, breaking down reasoning strategies with data-driven, personalized study plans tailored to each learner's needs; one student's score rose from **1100 to 1300** in a month

Groundskeeping Crew, The Groundsman – Augusta, GA — Apr. 2026

- Operated golf carts and transported equipment for The Masters Tournament, working fast-paced early-morning shifts to keep logistics running for daily crowds of thousands under tight time constraints

LEADERSHIP & AWARDS

CyberPatriot, Team Captain, Lakeside High & Stallings Island Middle School – Martinez, GA — 2020 – 2026

- Achieved Platinum Tier all **six** years in the Air Force Association's CyberPatriot competition, ranking **top 30%** at State by securing Windows/Linux systems and analyzing network traffic with Wireshark in timed, multi-round team competitions